

Econ 2 - Lecture 13 - 2/25 / 26

Lecture Quiz # 7 Released Today

↳ Due Monday, March 2nd

Last Class: Bad tech + Fiscal Policy

Today: Deficit & Debt = Chapter 5.1

$$\text{Expenditure Multiplier} = \frac{1}{1 - mpc}$$

$$\text{Tax Multiplier} = \frac{-mpc}{1 - mpc}$$

Fiscal Policy $\Rightarrow$ Difficult to implement

Great Financial Crisis $\Rightarrow$ Crisis $\Rightarrow \Delta Y = -1 \text{ Trillion}$
Action

ARRA / Stimulus Bill

$$\Delta G = 500 \text{ Bil}, \Delta T = -300 \text{ Bil}$$

$$\text{Low } mpc = ?, mpc = 0.5$$

$$\text{Expenditure Multiplier} = \frac{1}{1 - 0.5} = 2$$

$$\text{Tax Multiplier} = \frac{-0.5}{1 - 0.5} = -1$$

$$\left. \begin{array}{l} \Delta Y = 2 \times 500 \\ \Delta Y = -1 \times (-300) \end{array} \right\} = 1000 + 300 = 1,300 \text{ B} = 1.3 \text{ T}$$



COVID Fiscal Policy

March/April 2020 = \$2 Trillion
December 2020 = \$1 Trillion
March 2021 = \$2 Trillion

} = \$5 Trillion

Consequence → Government needs to borrow!

$$\text{Deficit} = G - T > 0$$

← All gov't spending

→ transfer payments

→ Interest on Debt

→ Tax Revenue

2020: $G > T \Rightarrow 3$ Trillion Deficit

2021: $G > T \Rightarrow 2.77$ Trillion

22: $G > T \Rightarrow 1.4T$

23: $G > T \Rightarrow 1.7T$

Normalized Deficit → Divide by GDP

Today $\frac{\text{Deficit}}{\text{GDP}} \sim 6\% \rightarrow$ only high ($> 3\%$)
↓ during recessions & wars
Low UE!

$$\text{Debt} = \text{Sum of deficits} = \sum (G - T)$$

3 Yr Example

$$\text{Debt} = \underbrace{(G_1 - T_1)}_{= 50} + \underbrace{(G_2 - T_2)}_{= 100} + \underbrace{(G_3 - T_3)}_{= -75}$$

$\downarrow$ Borrow \$50 in Yr 1 $\downarrow$ Borrow \$100 in Yr 2 $\downarrow$ Pay back \$75 in Yr 3

$$\text{Debt} = 50 + 100 - 75 = \$75$$

$\downarrow$ Stock Variable $\swarrow \downarrow \searrow$ Flow Variable

Facts about National Debt

$$\text{Current Debt} = \$38.7 \text{ T}$$

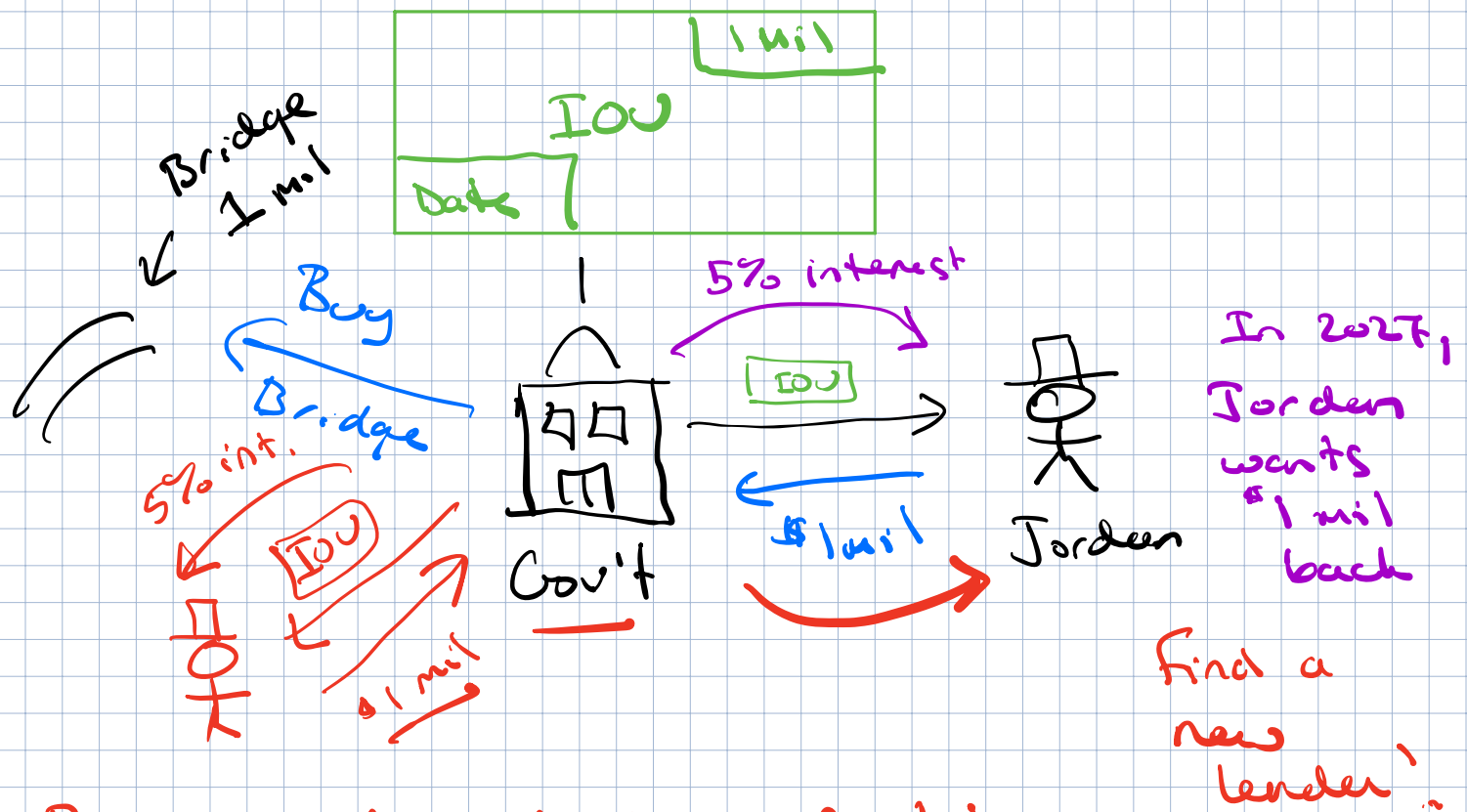
Fact #1: US Government DOES NOT owe \$38.7 to lenders

$$\text{Total Debt} = \underbrace{\text{Public Debt}}_{\substack{\text{Borrowing from outside the Gov't} \\ \text{(Bonds)} \\ \$31.1 \text{ T}}} + \underbrace{\text{Intra-Governmental Borrowing}}_{\substack{\text{Borrowing from within the Gov't} \\ \rightarrow \text{Social Security Trust, etc.} \\ = \$7.6 \text{ T}}}$$

Fact #2: We DO NOT need to pay back the \$3T in public debt?

Setting: Government borrows \$1 million from lenders
Government: vehicle for gov't to borrow money
Bond

"I owe you" (IOU) → 1 mil in one year



Roll over the debt indefinitely

- Concerns about paying interest
- Rolling debt over

Hard to find Lenders

As long as we believe they will be solvent

What do we need to worry about?

Interest Payments on debt

Where does the gov't generate money to pay interest? Tax Revenue!

Public Debt = \$1 million

Interest Rate = 5% = 0.05

Nominal GDP = \$5 million (Income today)

$$\begin{aligned}\text{Interest Payment} &= \text{Interest Rate} \times \text{Debt} \\ \text{On Debt} &= 0.05 \times 1 \text{ mil} = \$50,000\end{aligned}$$

As long as gov't can pay \$50,000 they can hold a debt of \$1 mil

→ Need to tax citizens → generate \$50,000

$$\text{Tax Revenue} \geq \text{Interest Payment}$$

$$\text{Tax Rate} \times \text{Nominal GDP} = \text{Interest Payment}$$

$$\text{Minimum Tax Rate} = \frac{\text{Interest Payment}}{\text{Nominal GDP}} \times 100$$

$$\begin{aligned}\text{Min. Tax Rate must be} &= \frac{50,000}{5,000,000} \times 100 = 1\% \\ 1\% \text{ in order to hold} & \\ \$1 \text{ mil in debt} &\end{aligned}$$

	Yr 1	Yr 2
Nom. GDP	200,000	300,000
Public Debt	100,000	200,000
Interest Rate	5%	5%
Price Level	100	120

What is the minimum tax rate?

Tax burden from interest on debt?

$$\begin{aligned}
 \text{Yr 1 Min. Tax Rate} &= \frac{\text{Int. Rate} \times \text{Debt}}{\text{Nom. GDP}} \times 100 \\
 &= \frac{5\% \times 100,000}{200,000} \times 100 \\
 &= 2.5\%
 \end{aligned}$$

$$\begin{aligned}
 \text{Yr 2 Min. Tax Rate} &= \frac{5\% \times 200,000}{300,000} \times 100 \\
 &= \frac{10,000}{300,000} \times 100 = \\
 &= 3.33\%
 \end{aligned}$$

Debt burden increased from 2.5% to 3.33%

What is the Real GDP in year 2? (Compare to Yr 1)

	Yr 1	Yr 2
Nom. GDP	200,000	300,000
Public Debt	100,000	200,000
Interest Rate	5%	5%
Price Level	100	120

(GDP Deflator)

$$\text{Price Level} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

$$\text{Real GDP} = \frac{\text{Nom. GDP}}{\text{Price Level}} \times 100$$

$$\text{Real GDP}_2 = \frac{300,000}{126} \times 100$$

$$\text{Real GDP}_2 = 250,000$$

Real GDP = 200,000

In yr 1: Real GDP = 200,000, Tax = 2.5%

In yr 2: Real GDP = 250,000, Tax = 3.33%
Was the tax increase worth it?

Fact #3: Debt ratio $\Rightarrow$ Appropriate
GDP measure
of Debt
Burden